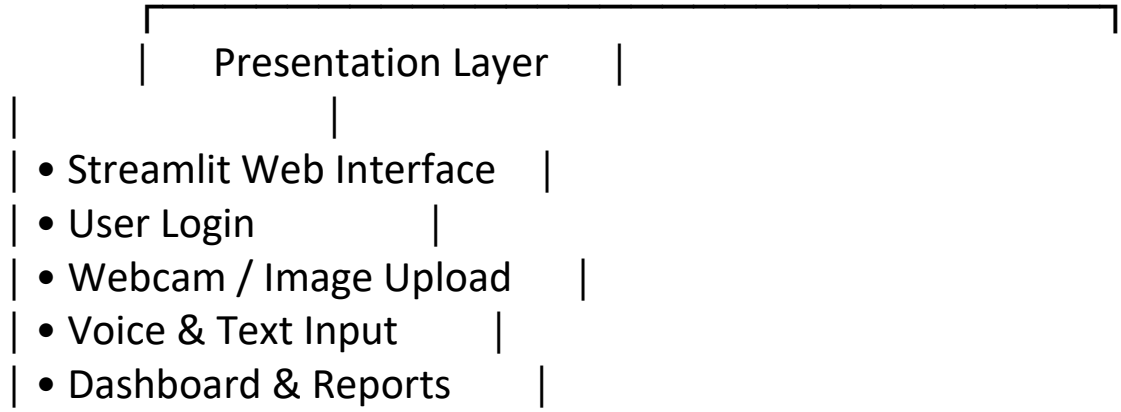
Project Demonstration

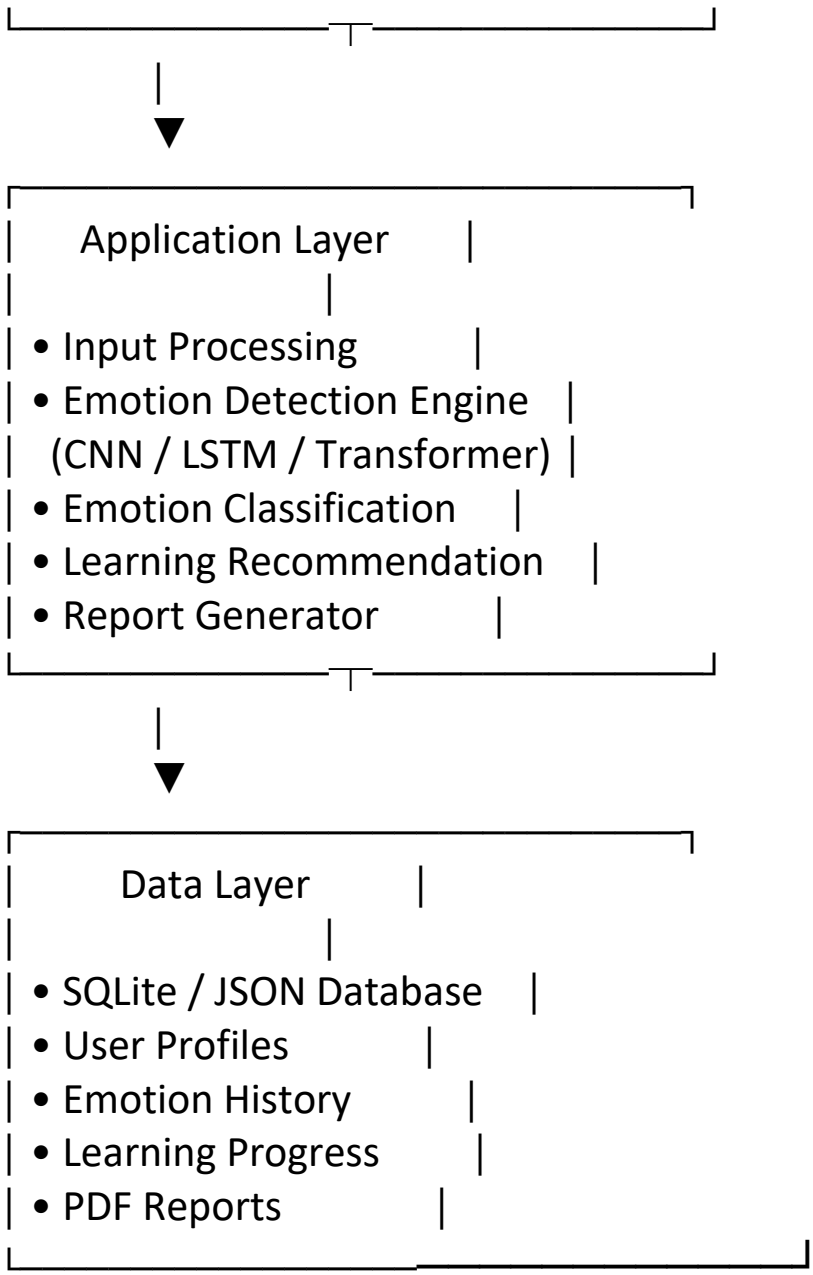
Date	4 July 2026
Team ID	XXXXXX
Project Name	Emotion Detection and Learning Support Engine
Maximum Marks	5 Marks

Project Demonstration Overview

This section demonstrates the complete workflow of the **Emotion Detection and Learning Support Engine (EDLSE)**. The application accepts **facial images, webcam input, voice recordings, or text**, detects the learner's emotional state using AI models, provides personalized learning recommendations, visualizes emotion analytics, and generates a comprehensive PDF report.

1. System Architecture




Application Layer
• Input Processing
• Emotion Detection Engine
(CNN / LSTM / Transformer)
• Emotion Classification
• Learning Recommendation
• Report Generator
Data Layer
• SQLite / JSON Database
• User Profiles
• Emotion History
• Learning Progress
• PDF Reports

2. Project Structure

```
Emotion-Detection-and-Learning-Support-Engine/
|
├── app.py
├── requirements.txt
├── README.md
├── .env.example
├── .gitignore
|
├── models/
|   ├── emotion_model.h5
|   ├── recommendation_model.pkl
|   └── labels.json
|
├── dataset/
|   ├── images/
|   ├── audio/
|   └── text/
|
├── utils/
|   ├── emotion_detection.py
|   ├── recommendation.py
|   ├── report_generator.py
|   ├── database.py
|   └── preprocessing.py
|
├── reports/
|
```

```
├── assets/
|   ├── logo.png
|   └── icons/
|
├── static/
|   ├── style.css
|   └── scripts.js
|
├── tests/
|   ├── test_emotion.py
|   ├── test_reports.py
|   └── test_database.py
|
└── screenshots/
    ├── home.png
    ├── dashboard.png
    └── report.png
```

3. Running the Application

Figure 3: Streamlit Server Started Successfully

Steps to Run

```
pip install -r requirements.txt
streamlit run app.py
```

The Streamlit server starts successfully and generates a local URL (for example, `http://localhost:8501`) to access the application.

4. Streamlit Application Running

Figure 4: Application Running on Localhost

The application launches successfully in the web browser, displaying the login page and the main dashboard for emotion detection and personalized learning support.

---

5. Application Workflow

Figure	Description
Figure 5	Home Page / Login Screen
Figure 6	Face Image / Voice / Text Input Page
Figure 7	Emotion Detection Interface (Webcam or Uploaded Input)
Figure 8	AI Emotion Detection Result (Detected Emotion & Confidence Score)
Figure 9	Personalized Learning Recommendations
Figure 10	Analytics Dashboard Showing Emotion History and Learning Progress
Figure 11	Generated PDF Report

6. Demonstration Summary

The Emotion Detection and Learning Support Engine (EDLSE) was successfully executed and tested. All major functionalities performed as expected during the demonstration, including:

- **User authentication and application launch.**
- **Face image, voice, and text input processing.**
- **AI-based emotion detection and emotion classification.**
- **Personalized learning recommendations based on detected emotions.**
- **Interactive dashboard displaying emotion trends and learning progress.**
- **Downloadable PDF report generation.**
- **Secure storage of user data and reports.**